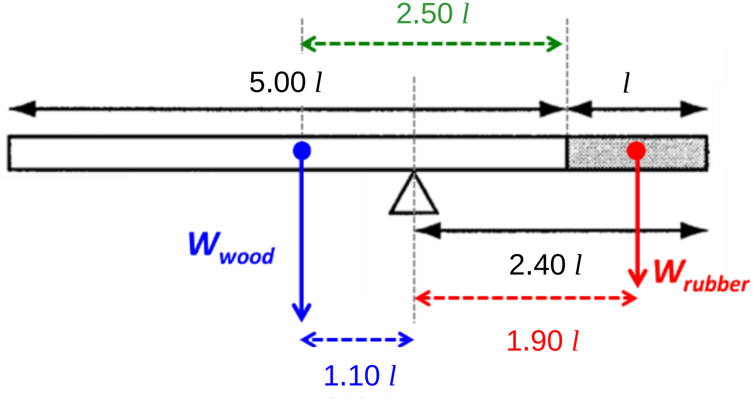
<table>
<tr><td></td><td></td><td>$\frac{}{dt} = \frac{}{v} = \frac{}{1000} = 4.9\ kg\ s$</td></tr>
<tr><td>7</td><td>C</td><td>



Taking moments about the CG (pivot) and letting A be the x-sectional area,

W_{wood} (distance of center-of-mass of wood from pivot) = W_{rubber} (distance of center-of-mass of rubber from pivot)

$A(5.00\ l)\rho_{wood}\, g(1.10\ l) = A(l)\rho_{rubber}\, g(1.90\ l)$

$5.00\ \rho_{wood}\,(1.10) = \rho_{rubber}\,(1.90)$

$\frac{\rho_{rubber}}{\rho_{wood}} = \frac{5.50}{1.90} = 2.89$

</td></tr>
</table>

8	D	[illegible]